

RISKFORGE AI

Credit Underwriting & Risk Intelligence Platform

NeoStats Candidate Submission Presentation

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1. Use Case & Problem Statement

PROBLEM STATEMENT:

- Class Imbalance: Default instances account for only ~8% of loan portfolios, biasing standard ML splits.
- Black-Box AI: Neural networks output scores without audit transparency, breaching regulations.
- Threshold Scales: Decision logic uses scaled coordinates rather than clean banker-friendly metrics.
- Static Reporting: Compliance auditors lack relational interfaces to query history dynamically.

RISKFORGE AI SOLUTION:

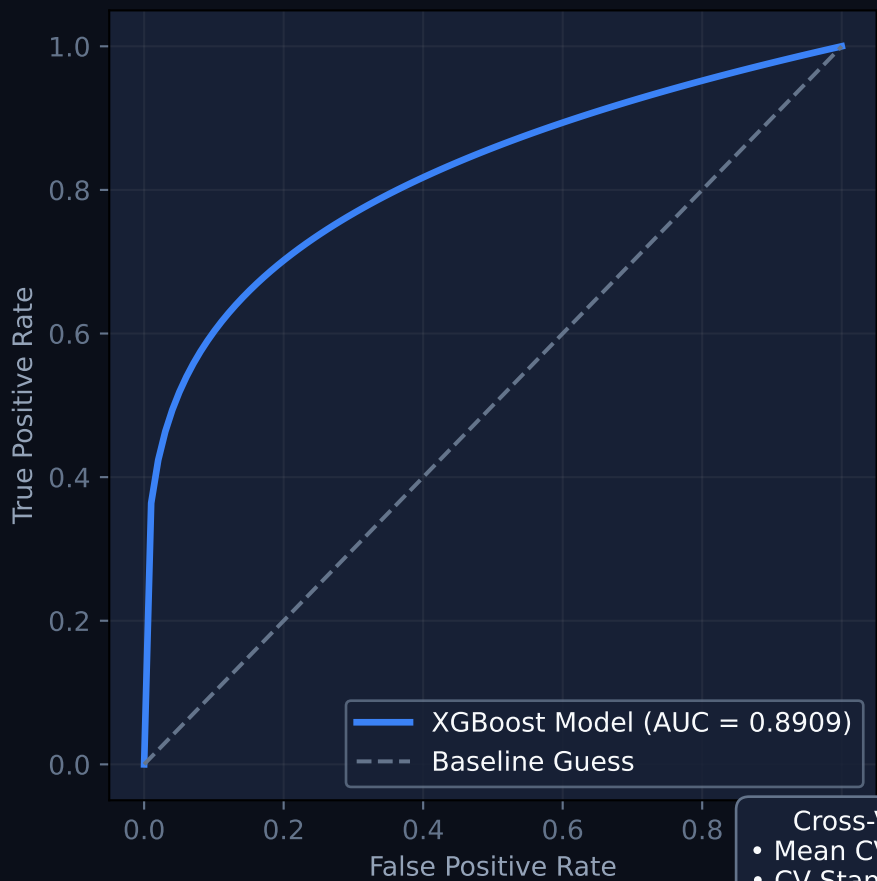
1. Stratified Validation: 5-Fold Stratified K-Fold XGBoost scoring avoiding target leakage.
2. SHAP Waterfall Audits: Complete breakdown of risk-increasing vs. mitigating drivers.
3. Banker-Friendly Rules: Extracted unscaled business rule sheets (e.g. Credit Score < 0.35).
4. Talk-to-Data Assistant: Conversational SQL interface querying SQLite with Gemini security checks.

2. System Architecture Flow



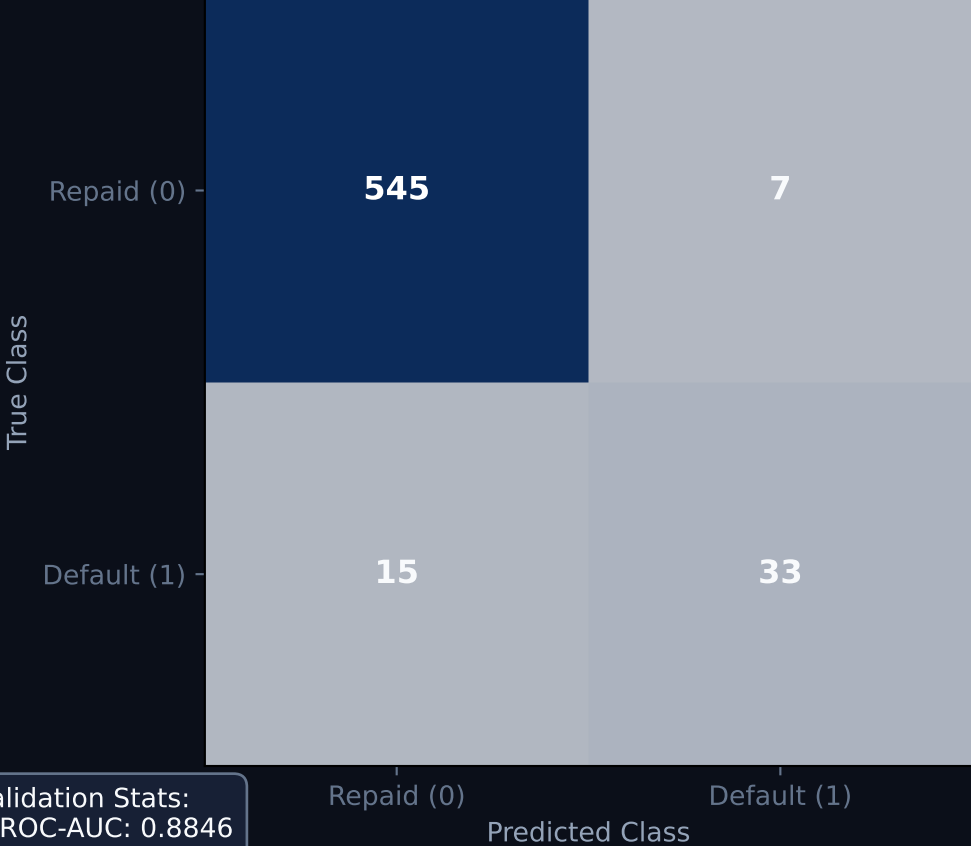
3. Machine Learning Performance Metrics

ROC Curve (Holdout Set)

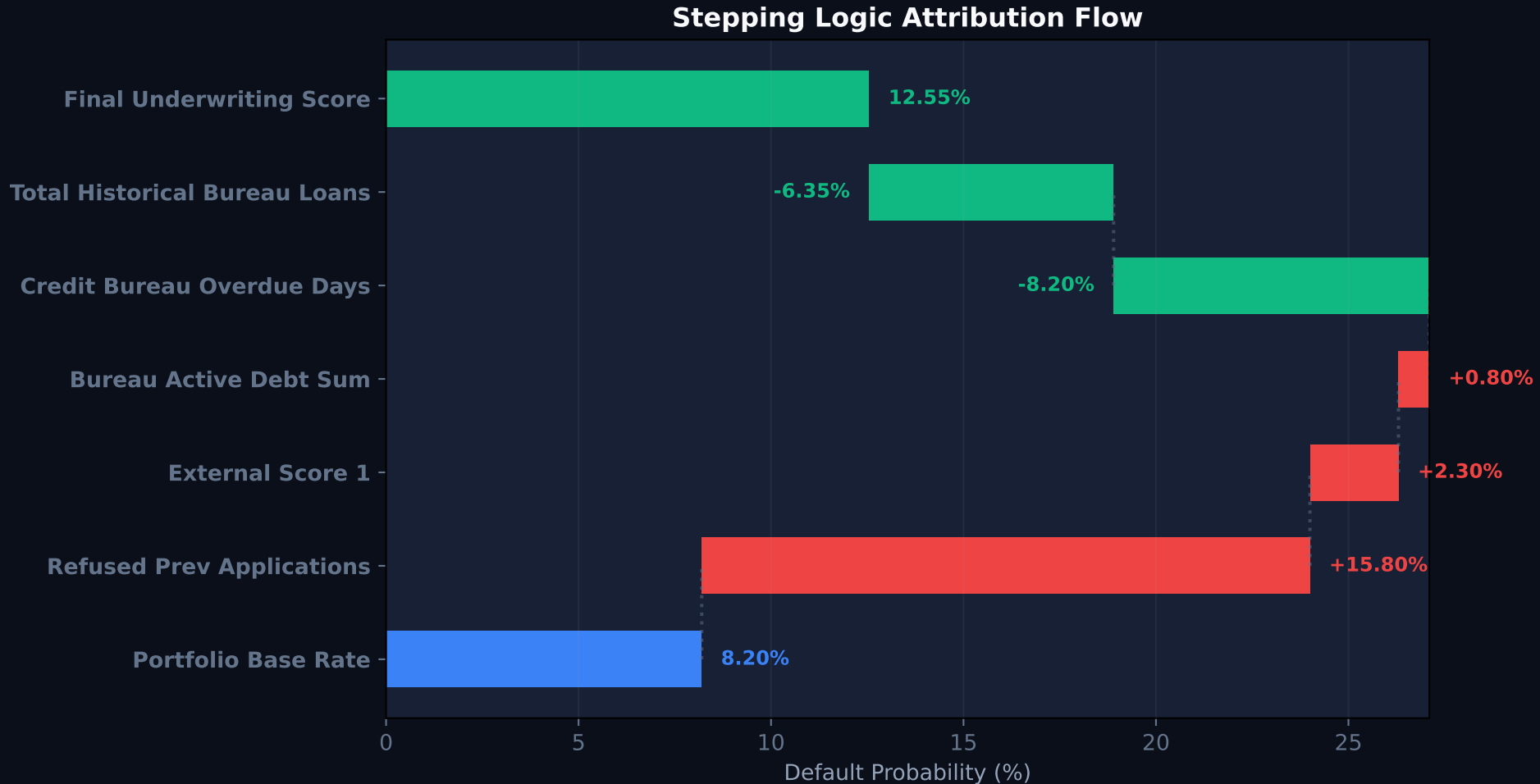


- Cross-Validation Stats:
- Mean CV ROC-AUC: 0.8846
 - CV Standard Dev: ± 0.0137
 - Holdout Accuracy: 96.33%
 - Holdout F1-Score: 0.7500

Confusion Matrix



4. SHAP Credit Underwriting Waterfall



5. Talk-to-Data Chatbot Engine

Analyst Question: 'How many customers defaulted?'

NL-to-SQL Processing Stream:

1. Prompt Mapping: Schema details mapped to system prompts.
2. Sanitization Check: Scans query for DROP, UPDATE, DELETE keywords.
3. SQLite Execution: Resolves database query locally against application_train.
4. Business Synthesis: Translates database rows back into plain English narrative.

Generated SQL Query:

```
SELECT COUNT(*) AS total_defaults FROM application_train WHERE TARGET = 1;
```

Execution Result Table:

total_defaults
240

Gemini Synthesized Response:

The database query identifies that a total of **240** applicants in the portfolio defaulted on their payment.